

**AI-BASED MULTICRITERIA OPTIMIZATION**

Introduction to Optimization

Lecture 2 – Tuesday May 15, 2018

L2, AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis 1

Outline

- Introduction to Optimization
- Optimization Problem Classifications
- Combinatorial Problems
- Optimization Methods

L2, AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis 2

Outline

- **Introduction to Optimization**
- Optimization Problem Classifications
- Combinatorial Problems
- Optimization Methods

Introduction to Optimization

Optimization is a term used to refer to a branch of computational science concerned with finding the “**best**” **solution** to a problem, possibly subject to a set of constraints.

“**best**” refers to **an acceptable (or satisfactory)** solution.

• Types of solutions

- ◇ A **feasible solution** satisfies all constraints.
- ◇ An **optimal solution** is feasible and provides the best objective function value.
- ◇ A **near-optimal solution** is feasible and provides a superior objective function value, but not necessarily the best.

Introduction to Optimization

• Statement of an Optimization Problem

An optimization or a mathematical programming problem can be stated as follows.

$$\text{Find } \mathbf{x} = \begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{Bmatrix} \text{ which minimize / maximize } f(\mathbf{X})$$

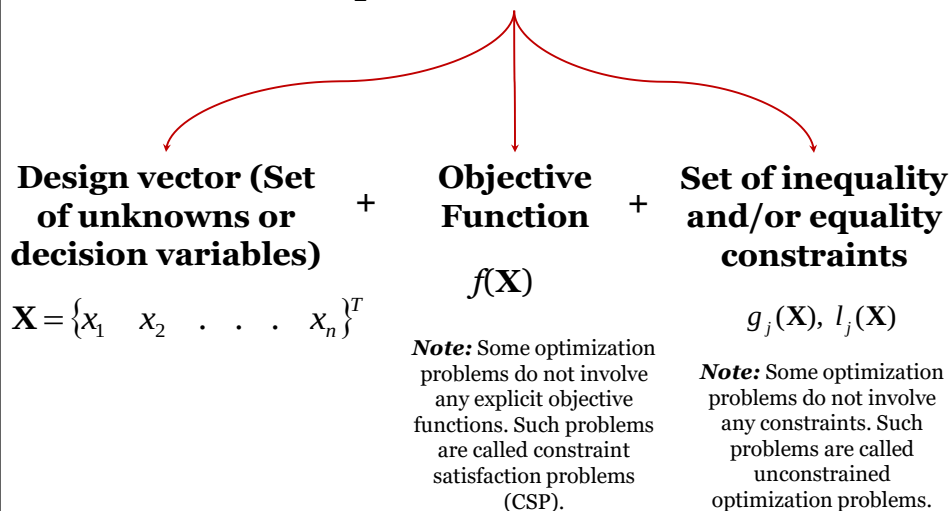
subject to the constraints

$$g_j(\mathbf{X}) \leq 0, \quad j = 1, 2, \dots, m \quad \text{Inequality constraints}$$

$$l_j(\mathbf{X}) = 0, \quad j = 1, 2, \dots, p \quad \text{Equality constraints}$$

Introduction to Optimization

Optimization Problem



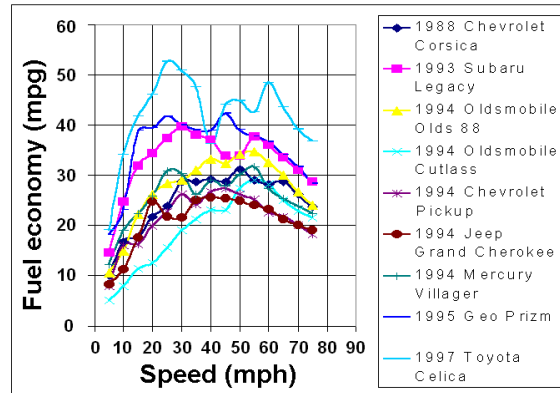
Introduction to Optimization

• Example-1:

◇ objective function

f =fuel economy

distance traveled per unit of fuel used; in miles per gallon (mpg) or kilometres per litre (km/L),



◇ Design vector or unknown

$\mathbf{X} = \{x_1\} = \{\text{car speed in mph or Km/h}\}$

◇ Constraints

$g_1(\mathbf{X}) = x_1 \leq \text{speed limit}$

Introduction to Optimization

• Example-2: Wheel Design

Find $\mathbf{x} = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{Bmatrix} = \begin{Bmatrix} \text{wheel diameter} \\ \text{breadth width} \\ \text{wheel loading} \\ \text{vehicle speed} \\ \text{contact area} \end{Bmatrix}$ which optimize $f = \begin{Bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \\ f_5 \end{Bmatrix} = \begin{Bmatrix} \text{wheel sinkage} \\ \text{motion resistance} \\ \text{drive torque} \\ \text{drive power} \\ \text{slope negotiability} \end{Bmatrix}$

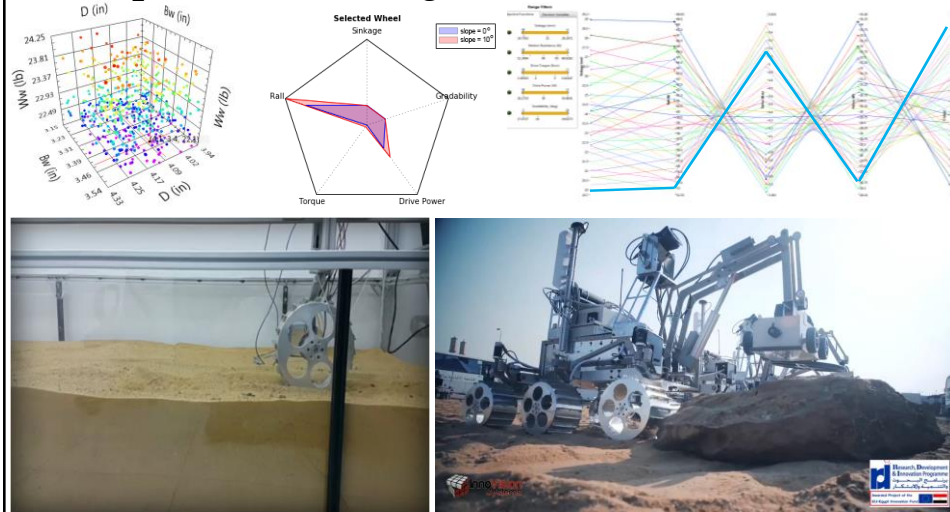
s.t. boundary constraints



For more info: Mohammed Ashraf and Alaa Khamis, "Design of Robot Wheels for Rough Terrain - A Multi-criteria Optimization Approach", Second International Workshop on Recent Advances in Robotics and Sensor Technology for Humanitarian Demining and Counter-IEDs (RST 2016).

Introduction to Optimization

• *Example-2: Wheel Design*



MineProbe: A Distributed Mobile Sensor System for Minefield Reconnaissance and Mapping: <http://mineprobe.org/>

L2, AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis

10

Introduction to Optimization

• **Design Vector/Decision Variables**

A set of unknowns or decision variables which affects the value of the objective function.

If $\mathbf{x}$ represents the unknowns, also referred to as the independent variables, then $f(\mathbf{x})$ **quantifies the quality** of the **candidate solution** or **feasible solution**, $\mathbf{x}$.

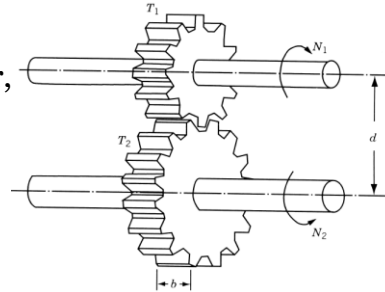
L2, AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis

11

Introduction to Optimization

• Design Vector: Example

- ◇ Consider the design of the gear pair, characterized by its face width b , number of teeth T_1 and T_2 , center distance d , pressure angle ψ , tooth profile, and material.



- ◇ If center distance d , pressure angle ψ , tooth profile, and material of the gears are fixed in advance, these quantities can be called **preassigned parameters**.
- ◇ The remaining quantities can be collectively represented by a **design vector** $\mathbf{X} = \{x_1, x_2, x_3\}^T = \{b, T_1, T_2\}^T$.

Ref.: Singiresu S. Rao. *Engineering Optimization: Theory and Practice*. John Wiley & Sons, 2009.

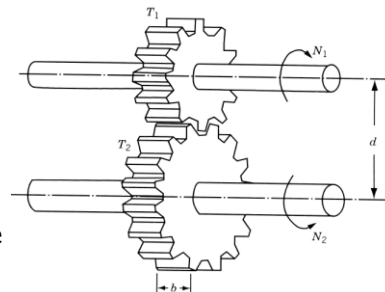
L2. AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis

12

Introduction to Optimization

• Design Vector Space

- ◇ In the case of the design of a gear pair,
- ◇ The design point $\{1.0, 20, 40\}^T$, for example, represents a **possible solution (design point)**,
- ◇ Whereas the design point $\{1.0, -20, 40.5\}^T$ represents an **impossible solution**
- ◇ since it is not possible to have either a negative value or a fractional value for the number of teeth.



Ref.: Singiresu S. Rao. *Engineering Optimization: Theory and Practice*. John Wiley & Sons, 2009.

L2. AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis

14

Introduction to Optimization

• Set of constraints

A set of constraints that **restricts** the values that can be assigned to the unknowns.

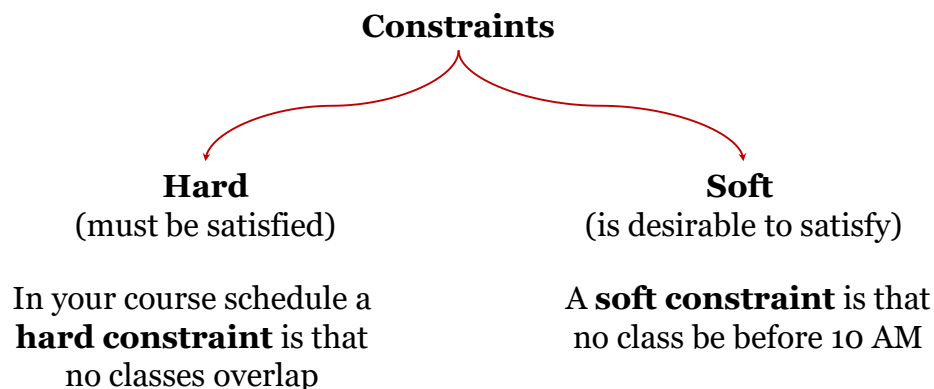
Most problems define at least a set of **boundary constraints**, which define the domain of values for each variable.

Constraints can, however, be more **complex**, **excluding certain candidate solutions** from being considered as solutions.

“Constrained optimization is the art of compromise between conflicting objectives. This is what design is all about.” William A. Dembski, Signs of Intelligence: Understanding Intelligent Design

Introduction to Optimization

• Set of constraints (cont'd)



Introduction to Optimization

• Set of constraints (cont'd)

Constraints

Explicit

(are explicitly stated in a given problem)

Implicit

(are part of the natural description of the phenomenon under study).

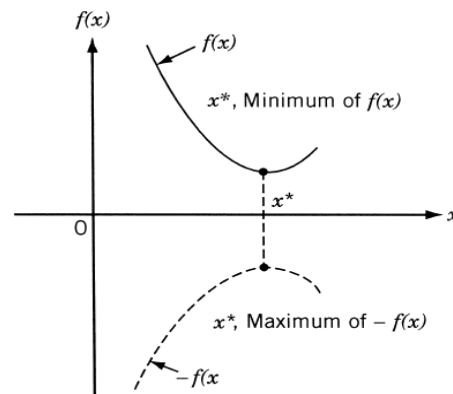
These are typically bound constraints on the decision variables
For example, the **width of an object is necessarily non-negative**.

Introduction to Optimization

• Objective Function

An objective function (also known as the criterion or merit function) represents the quantity to be optimized, that is, the quantity to be minimized or maximized.

- ◇ If a point x^* corresponds to the **minimum value** of function $f(x)$,
- ◇ the same point also corresponds to the **maximum value** of the negative of the function, $-f(x)$.

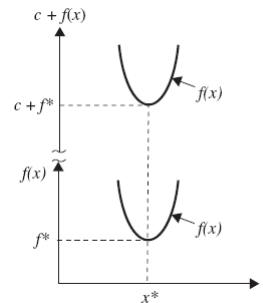
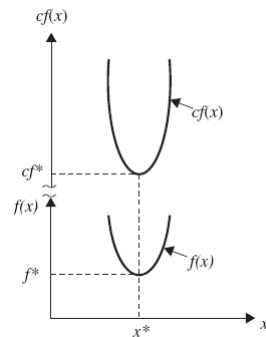


Introduction to Optimization

• Objective Function

◇ The following operations on the objective function will not change the optimum solution x^* .

- Multiplication (or division) of $f(x)$ by a positive constant c .
- Addition (or subtraction) of a positive constant c to (or from) $f(x)$.



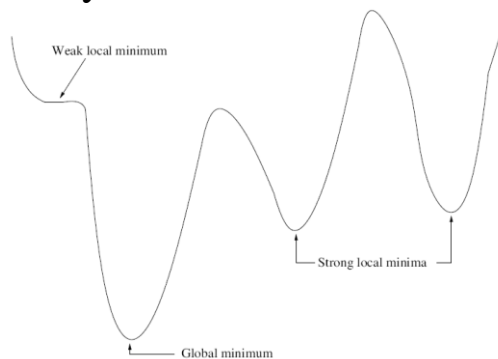
L2. AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis

22

Introduction to Optimization

• Objective Function: Optimality Conditions

◇ **Global Optimum:** a global optimum of global minimum (considering a minimization problem) is formally defined as follows:



Global minimum: The solution $\mathbf{x}^* \in \mathcal{F}$, is a global optimum of the objective function, f , if

$$f(\mathbf{x}^*) < f(\mathbf{x}), \forall \mathbf{x} \in \mathcal{F}$$

where $\mathcal{F} \subseteq \mathcal{S}$.

Ref.: A. Engelbrecht. Fundamentals of Computational Swarm Intelligence. Wiley, 2005

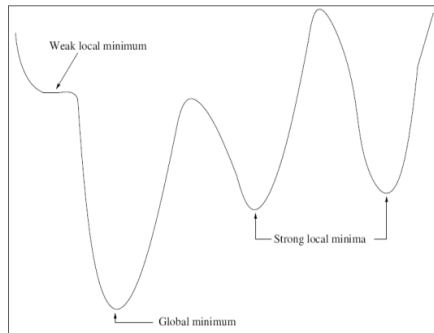
L2. AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis

23

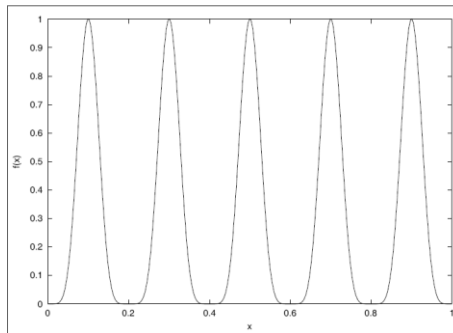
Introduction to Optimization

• Objective Function: Optimality Conditions

The global optimum is therefore the best of a set of candidate solutions.



Global optimum for a minimization problem



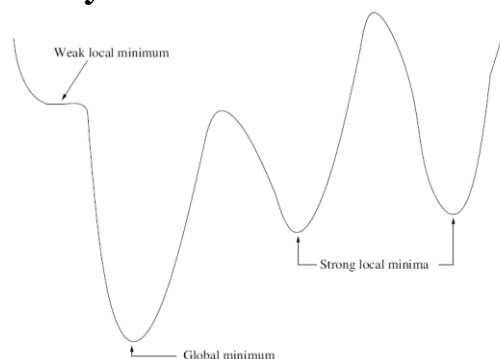
A problem may have more than one global optimum.

$$f(x) = \sin^6(5\pi x)$$

Introduction to Optimization

• Objective Function: Optimality Conditions

- ◇ **Strong Local Minimum:** A strong local minimum is defined as follows:



Strong local minimum: The solution, $\mathbf{x}_{\mathcal{N}}^* \in \mathcal{N} \subseteq \mathcal{F}$, is a strong local minimum of f , if

$$f(\mathbf{x}_{\mathcal{N}}^*) < f(\mathbf{x}), \forall \mathbf{x} \in \mathcal{N}$$

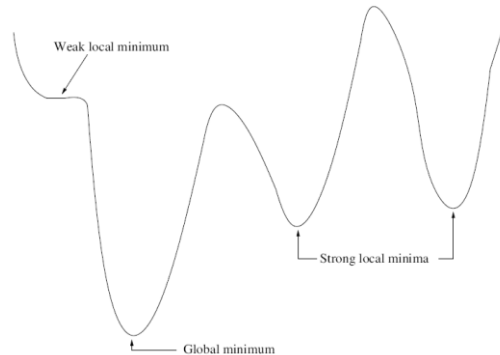
where $\mathcal{N} \subseteq \mathcal{F}$ is a set of feasible points in the neighborhood of $\mathbf{x}_{\mathcal{N}}^*$.

Introduction to Optimization

• Objective Function: Optimality Conditions

◇ Weak Local

Minimum: A weak local minimum is defined as follows:



Weak local minimum: The solution, $\mathbf{x}_N^* \in \mathcal{N} \subseteq \mathcal{F}$, is a weak local minimum of f , if

$$f(\mathbf{x}_N^*) \leq f(\mathbf{x}), \forall \mathbf{x} \in \mathcal{N}$$

where $\mathcal{N} \subseteq \mathcal{F}$ is a set of feasible points in the neighborhood of $\mathbf{x}_N^*$.

Introduction to Optimization

• Objective Function

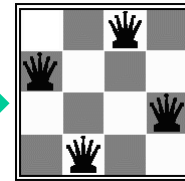
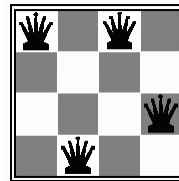
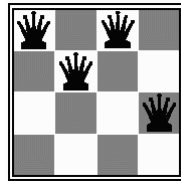
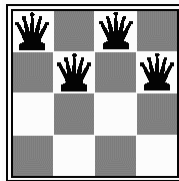
Some problems, specifically **constraint-satisfaction problems (CSP)**, do not define an explicit objective function. Instead, the objective is to find a solution which satisfies all of a set of constraints.

Introduction to Optimization

• Objective Function: Constraint-Satisfaction Problems

◇ Example: N-queen problem

Put n queens on an $n \times n$ board with no two queens on the same row, column, or diagonal



Move a queen to reduce number of conflicts

Goal configuration

no queen is
attacking
another queen

Introduction to Optimization

• Objective Function: Constraint-Satisfaction Problems

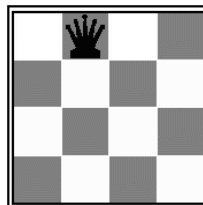
◇ Example: N queen problem

Assume:

each queen is represented as a row: Q1, Q2, Q3, and Q4

An assignment consists of assigning a column to a queen

$\{ Q1 = 2 \} \rightarrow$ places a queen in row 1, column 2



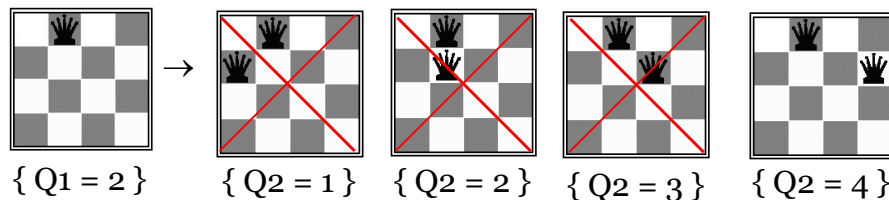
Introduction to Optimization

• Objective Function: Constraint-Satisfaction Problems

◇ Example: N queen problem

The constraints on the problem restrict certain values for each variable so that all assignments are consistent.

For example, after we have assigned Q_1 , and now want to assign Q_2 , we know we cannot use value 2, since this would violate a constraint: Q_1 could attack Q_2 and visa versa.



L2, AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis

30

Introduction to Optimization

• Objective Function: Constraint-Satisfaction Problems

◇ Example: N queen problem

Variables: $\{ Q_1, Q_2, Q_3, Q_4 \}$

Domain: $\{ (1, 2, 3, 4), (1, 2, 3, 4), (1, 2, 3, 4), (1, 2, 3, 4) \}$

Constraints: *Alldifferent*(Q_1, Q_2, Q_3, Q_4) and

for $i = 0 \dots n$ and $j = (i+1) \dots n$, $k = j-i$,

$Q[i] \neq Q[j] + k$ and

$Q[i] \neq Q[j] - k$.

Ref.: CSP Tutorial, Cork Constraint Computation Centre (4C): <http://4c.ucc.ie/web/outreach/index.html>. Last accessed: Feb. 7, 2017

L2, AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis

31

Introduction to Optimization

• Objective Function

An optimization problem involving **multiple objective functions** is known as a **multiobjective optimization problem**.

With multiple objectives there arises a **possibility of conflict**, and one simple way to handle the problem is to construct an **overall objective function** as a linear combination of the conflicting multiple objective functions.

$$f(X) = \alpha_1 f_1(X) + \alpha_2 f_2(X)$$

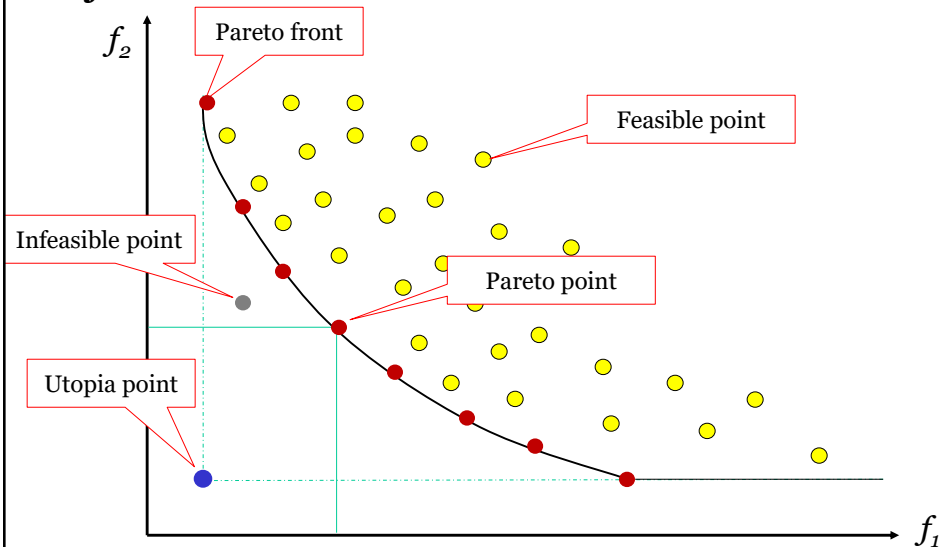
where α_1 and α_2 are constants whose values indicate the relative importance of one objective function relative to the other.

L2, AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis

32

Introduction to Optimization

• Objective Function



L2, AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis

33

Introduction to Optimization

- **Example-1: Graphical Optimization**

$$\mathbf{X} = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} \quad f(\mathbf{X}) = x_1 + x_2 \quad \begin{aligned} g_1(\mathbf{X}) &= 3x_1 - x_2 \geq 3 \\ g_2(\mathbf{X}) &= x_1 + 2x_2 \leq 5 \\ g_3(\mathbf{X}) &= x_1 + x_2 \leq 4 \\ g_4(\mathbf{X}) &= x_1 \geq 0, x_2 \geq 0 \end{aligned}$$

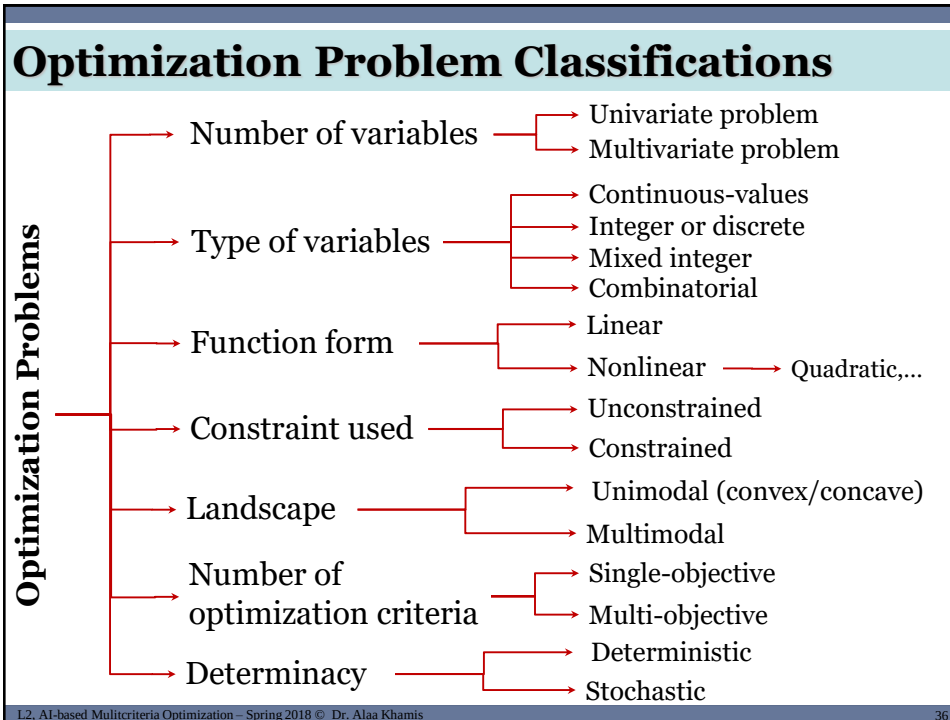
$$\text{find } \mathbf{X}^* = \arg \min_{\mathbf{X}} f(\mathbf{X})$$

$$\text{s.t. } g_i(\mathbf{X}) \quad i = 1, 2, 3, 4$$

Solution: TBD on the whiteboard...

Outline

- Introduction to Optimization
- **Optimization Problem Classifications**
- Combinatorial Problems
- Optimization Methods



Optimization Problem Classifications

- **Number of variables**

The number of variables that influences the objective function:

A problem with **only one variable** to be optimized is referred to as a **univariate problem**.

If more than one variable is considered, the problem is referred to as a **multivariate problem**.

L2, AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis 37

Optimization Problem Classifications

- **The type of variables**

A **continuous problem** has continuous-valued variables, i.e. $x_j \in \mathbb{R}$, for each $j = 1, \dots, n_x$

If $x_i \in \mathbb{Z}$, the problem is referred to as an **integer or discrete optimization** problem.

A **mixed integer problem** has both continuous-valued and integer-valued variables.

Problems where solutions are **permutations of integer-valued** variables are referred to as **combinatorial optimization problems** (finding an optimal object from a finite set of objects).

Optimization Problem Classifications

- **Function form**

Linear problems have an objective function which is linear in the variables.

Quadratic problems use quadratic functions.

When other nonlinear objective functions are used, the problem is referred to as a **nonlinear problem**.

Optimization Problem Classifications

- **The constraints used**

A problem which uses only **boundary constraints** is referred to as an **unconstrained problem**.

Constrained problems have additional **equality and/or inequality constraints**.

Optimization Problem Classifications

- **The constraints used**

Converting **constrained optimization to unconstrained optimization** using Lagrange multiplier

Find $\mathbf{x}$ which minimize /maximize $f(\mathbf{x})$

s.t. $x > a$



Find $\mathbf{x}$ which minimize /maximize $L = f(\mathbf{x}) - \sum \alpha(x - a)$

$\partial f(\mathbf{x}) = 0, \partial(x - a) = 0$

where α is the Lagrange multiplier



Joseph-Louis Lagrange
(1736-1813)
Italian mathematician
and astronomer

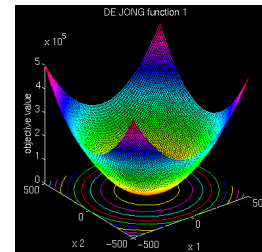
Optimization Problem Classifications

- **Landscape**

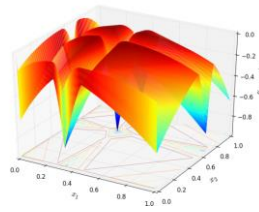
If there exists only one clear solution, the problem is **unimodal**.

If more than one optimum exists, the problem is **multimodal**.

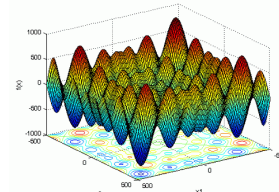
Some problems may have false optima, in which case the problem is referred **deceptive**.



Unimodal or Convex function



Deceptive function



Multimodal function

Optimization Problem Classifications

- **The number of optimization criteria**

If the quantity to be optimized is expressed using **only one objective function**, the problem is referred to as a **mono-objective** or **uni-objective problem**.

A **multi-objective problem** specifies more than one sub-objective which need to be simultaneously optimized.

Optimization Problem Classifications

- **Determinacy**

All the optimization problems are **deterministic** in the sense that for any given set of design variables, the values of both **objective functions and the constraint functions are determined exactly**.

In the **real world**, we can only know some parameters to a certain extent with uncertainty. Thus, if there is any **uncertainty and noise in the design variables and the objective functions and/or constraints**, then the optimization becomes a **stochastic optimization problem**, or a robust optimization problem with noise.

Optimization Problem Classifications

- **Examples:** State the type of the following optimization problems:

a) $\max \exp(-x^2)$ subject to $-2 \leq x \leq 5$;

Constrained nonlinear optimization with a single objective (uni-objective).

Optimization Problem Classifications

- *Examples (cont'd):*

b) $\min x^2 + y^2, (x-2)^2, (y-3)^3$

Unconstrained multiobjective optimization.

Optimization Problem Classifications

- *Examples (cont'd):*

c) Design a quieter, more efficient, low cost car engine;

Multiobjective optimization with multiple constraints.

Optimization Problem Classifications

- **Simple vs. hard optimization problems**

Simple Problems	Hard Problems
Few decision variables	Many decision variables
Differentiable	Discontinuous, combinatorial
Unimodal	Multimodal
Objective easy to calculate	Objective difficult to calculate
No or light constraints	Severely constraints
Feasibility easy to determine	Feasibility difficult to determine
Single objective	Multiple objective
Deterministic	Stochastic

L2, AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis

48

Outline

- Introduction to Optimization
- Optimization Problem Classifications
- **Combinatorial Problems**
- Optimization Methods

L2, AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis

49

Combinatorial Problems

Combinations and Permutations: what is the difference?

- If the order doesn't matter, it is a Combination.
- If the order does matter it is a Permutation.
- A Permutation is an ordered Combination.



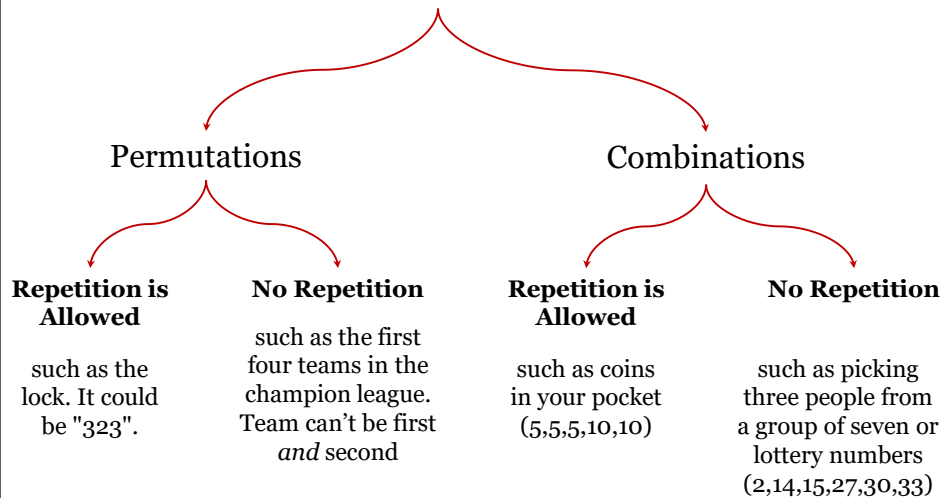
The berry fruit salad is a combination of strawberries, blueberries and blackberries .



The combination to the safe was 472

Combinatorial Problems

Combinatorics



Combinatorial Problems

Combinatorial problems are class of problems in which the **decision variables** are discrete -i.e. where the **solution** is a **set (combination)**, or a **sequence (permutation)**, of **integers** or other **discrete objects**.

Ref.: C. Reeves, Modern Heuristic Techniques for Combinatorial Problems. Halsted Press, New York, 1993

L2, AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis

59

Combinatorial Problems

• Examples: TSP

Given n cities, a travelling salesman must visit the n cities and return home, making a loop (roundtrip). He would like to travel in the most efficient way (cheapest way or shortest distance or some other criterion).

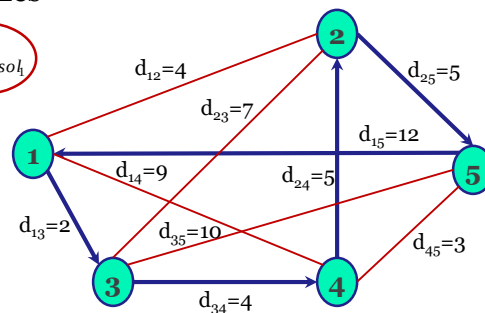
Find the route which minimizes

$$total_dist = \sum_{i=1}^{n-1} d_{sol_i sol_{i+1}} + d_{sol_n sol_1}$$

For a closed tour

The solution represented by permutation

Sol = [1, 3, 4, 2, 5]



L2, AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis

60

Combinatorial Problems

• Examples: Knapsack problem

Given n items, each with a weight w_i and a value u_i , to be packed into a bag of capacity C .

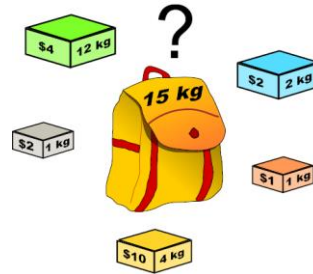
Determine the subset I of items which should be packed in order to maximize

$$\sum_{i \in I} u_i$$

such that

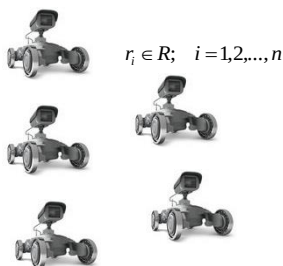
$$\sum_{i \in I} w_i \leq C$$

The solution is represented by a combination



Combinatorial Problems

• Examples: Task Allocation Problem



A set of n mobile Sensors: R



A set of m surveillance tasks: T

A set of n mobile sensors to carry out m surveillance tasks. If the sensor i does task j , it costs c_{ij} units. Find an assignment π which minimizes

$$\sum_{i=1}^n \sum_{j=1}^m c_{ij} \pi_{ij}$$

The solution represented by combination of pairs

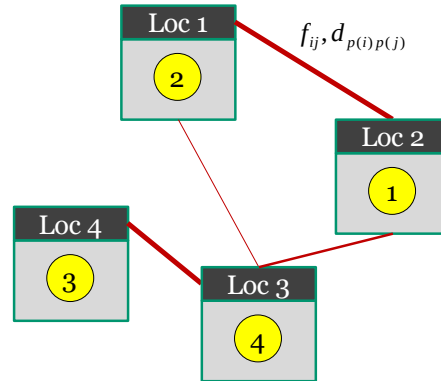
Combinatorial Problems

• Examples: Plant Layout Problem (PLP)

Given n facilities (departments) and n locations, PLP aims at assigning different facilities (departments) to different locations in order to minimize the total material handling cost.

$$TMHC = \sum d_{ij} \times f_{ij}$$

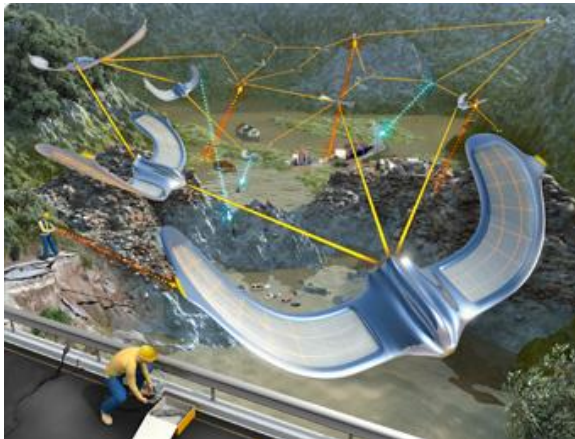
The solution is represented by a combination of pairs.



Combinatorial Problems

• Examples: Communication Relaying

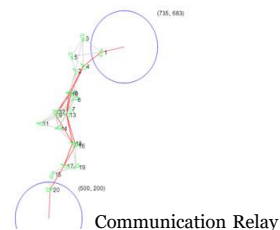
- ◇ Unmanned Aerial/Ground/Marine Vehicles (UXVs)



UAVs Deployment

Maximize:

- Area Coverage
- Object Coverage
- Radio Coverage



Outline

- Introduction to Optimization
- Optimization Problem Classifications
- Combinatorial Problems
- **Optimization Methods**

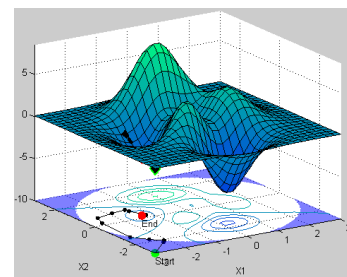
L2, AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis

65

Optimization Methods

- **Classifications based on type of solution that is located**

◇ **Local Search:** uses only local information of the search space surrounding the current solution to produce a new solution. Since only local information is used, local search algorithms (**local optimizer**) locate **local optima** (which may be a global minimum).



◇ **Global Search:** uses more information about the search space to locate a **global optimum**. It is said that **global search algorithms** explore the **entire search space**, while **local search algorithms** exploit **neighborhoods**.

L2, AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis

69

Optimization Methods

- **Classifications based on optimization speed**

- ◇ **Online Optimization:** handle tasks that need to be solved quickly in a time span between **ten milliseconds** to **a few minutes**. In order to find a solution in this short time, optimality is normally **traded in for speed gains**.

Examples: robot localization, load balancing, services composition for business processes, or updating a factory's machine job schedule after new orders came in.

From the examples, it becomes clear that online optimization tasks are often carried out **repetitively** – new orders will, for instance, continuously arrive in a production facility and need to be scheduled to machines in a way that minimizes the waiting time of all jobs.

Optimization Methods

- **Classifications based on optimization speed**

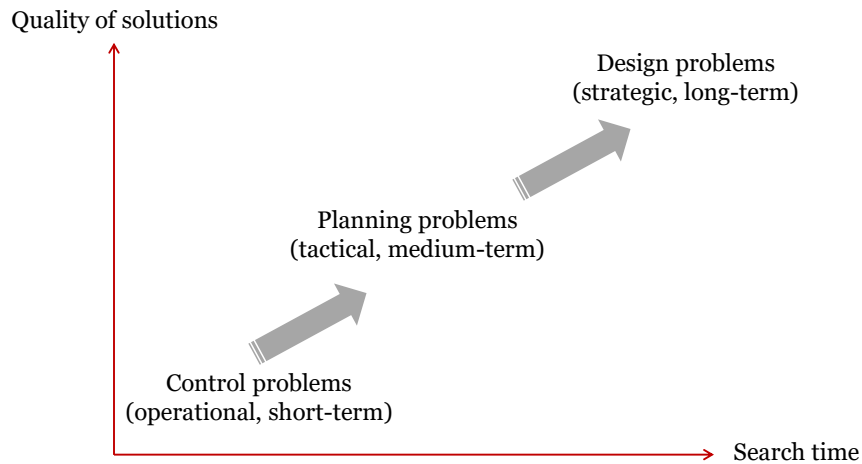
- ◇ **Offline Optimization:** time is not so important and a user is willing to wait maybe even days if he/she can get an optimal or close-to-optimal result.

Such problems regard for example **design optimization**, **data mining**, or creating **long-term schedules for transportation crews**.

These optimization processes will usually be **carried out only once** in a long time.

Optimization Methods

• Classifications based on optimization speed



Different classes of problems in terms of the trade-off between quality of solutions and search time: design (strategic, long-term), planning (tactical, medium-term), control (operational, short-term).

Ref.: El-Ghazali Talbi. Metaheuristics: From Design to Implementation. Wiley, ISBN: 978-0-470-27858-1

L2. AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis

72

Optimization Methods

• Other Classifications

◇ Exact Methods

- Find the optimal solution,
- High computational costs.

◇ Approximate Methods

- Find sub-optimal solutions,
- Low computational costs.

L2. AI-based Multicriteria Optimization – Spring 2018 © Dr. Alaa Khamis

73

Optimization Methods

• Our Classification

